



MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO
DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 02



**To design and analyze the behavior of AND, OR, NOT Gates using
Switches, Diodes, Transistor**

Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGN AND CODING	OBSERVATION/ RESULTS	SCORE

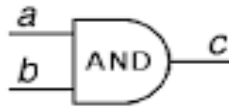
Objective: To design and analyze the behavior of AND, OR, NOT Gates using Switches, Diodes, Transistor

Apparatus:

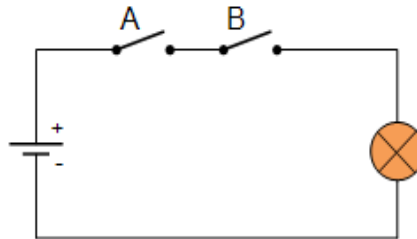
- Diodes
- Transistors
- Resistors (10K ohm)
- Switches (SPST)
- Breadboard
- Multimeter
- Connecting wire

Theory:

A **Logic AND Gate** is a type of digital logic gate that has an output which is normally at logic level "0" and only goes "HIGH" to a logic level "1" when ALL of its inputs are at logic level "1". The output of a **Logic AND Gate** only returns "LOW" again when ANY of its inputs are at a logic level "0". The logic or Boolean expression given for a logic AND gate is that for *Logical Multiplication* which is denoted by a single dot or full stop symbol, (.) giving us the Boolean expression of $A.B = Q$.

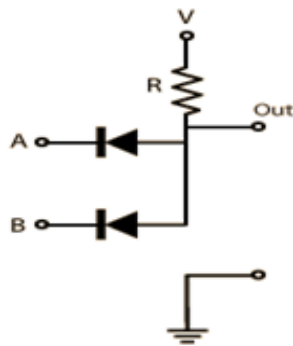


Logic Diagram

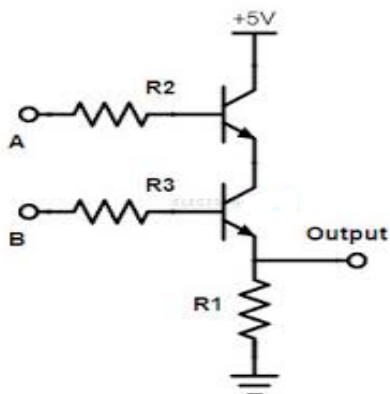


Switch A - Open = "0", Closed = "1"
Switch B - Open = "0", Closed = "1"

AND Gate using Switches



AND Gate using Diodes



AND Gate using Transistors

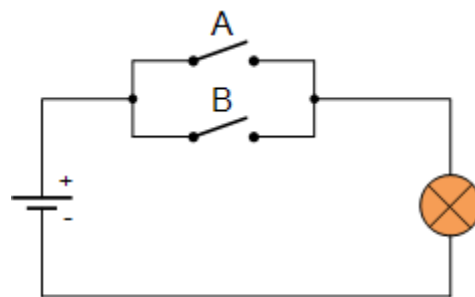
OR Gate:

The Logic OR Function states that an output action will occur or become TRUE if either one "OR" more events are TRUE, but the order at which they occur is unimportant as it does not affect the final result. For example, $A + B = B + A$. The logic or Boolean expression given for a logic OR gate is that for Logical Addition which is denoted by a plus sign, (+). Thus a 2-input (A,B) Logic OR Gate has an output term represented by the Boolean expression of $A+B = C$.



Logic Diagram

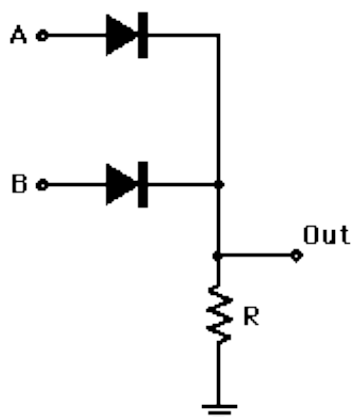
OR gate can be formed by using switches, diodes and transistors, circuit diagrams are given below.



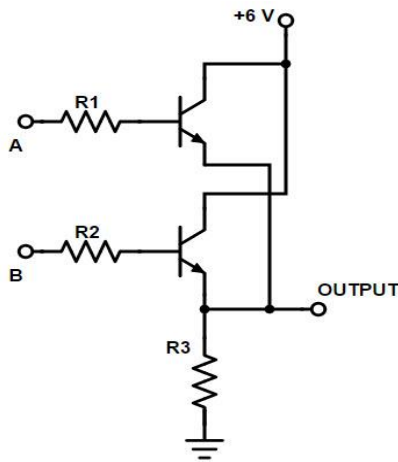
Switch A - Open = "0", Closed = "1"

Switch B - Open = "0", Closed = "1"

OR Gate using Switches



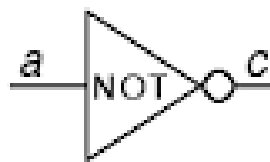
OR Gate using Diodes



OR Gate using Transistors

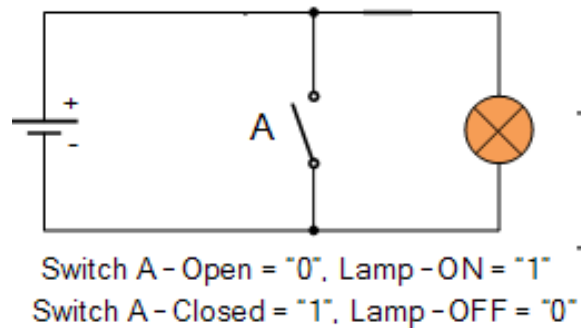
NOT Gate:

The Logic NOT Function is simply a single input inverter that changes the input of a logic level "1" to an output of logic level "0" and vice versa. The logic NOT function is so called because its output state is NOT the same as its input state. It is generally denoted by a bar or over line ($\bar{}$) over its input symbol which denotes the inversion operation. As NOT gates perform the logic INVERT or COMPLEMENTATION function they are more commonly known as Inverters because they invert the signal. In logic circuits this negation can be represented by a normally closed switch.

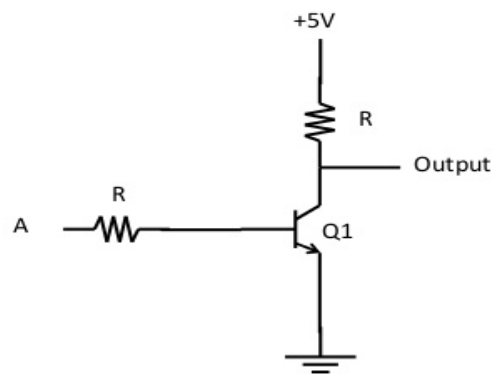


Logic Diagram

NOT gate can be formed by using switches, and transistors, circuit diagrams are given below.



NOT Gate using Switch



NOT Gate using Transistor

Voltage transfer characteristics:

One of the main properties of any digital circuit is the voltage transfer Characteristics which relates the output voltage to the input voltage under static conditions. The three regions of operation for the output transistor (cutoff, active and saturation) can be easily identified on the curve. The logic level, V_{OL} , V_{OH} , V_{IL} , and V_{IH} can be found from these characteristics as shown in figure.

Figure shows the transfer curve for TTL inverter without any fan-out. When the input voltage is 0 V, the output is HIGH at 3.3 V. As the input voltage is increased from 0 to 0.7 V, the output remains relatively constant (Region I). Beyond 0.7 V to about 1.2 V, the output decreases more gradually with increasing input voltage (Region II). The threshold voltage, the voltage on the transfer curve at which $V_{OUT} = V_{IN}$ and occurs in Region III, is found at the intersection of the transfer curve and the line $V_{OUT} = V_{IN}$. Finally, in Region IV, the output remains constant at 0.2 V as the input voltage is increased.

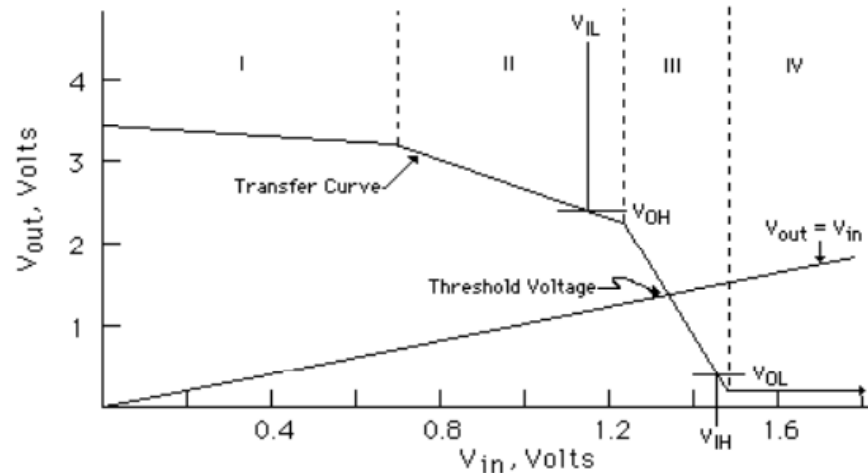


Figure : Voltage Transfer Curve of TTL inverter

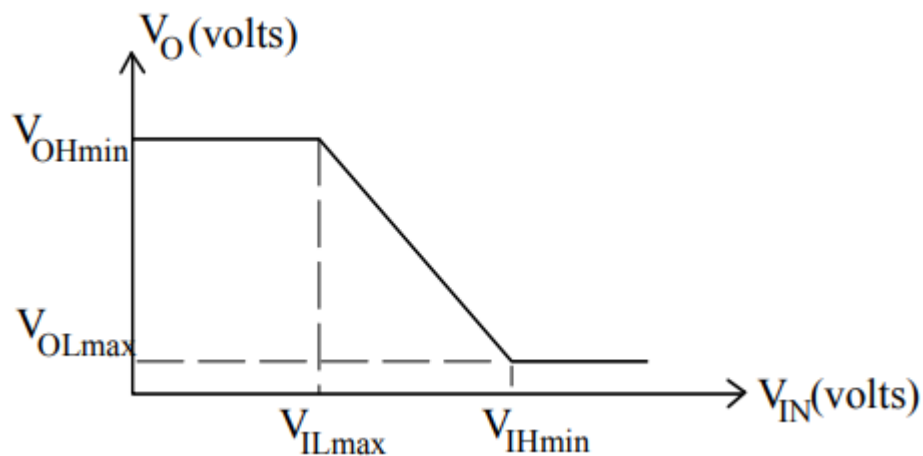


Figure : Voltage Transfer Curve of TTL inverter

Questions:

1. what is the purpose of diodes to implement AND logic and OR logic?

The basic purpose of diode in which used AND and OR gate .Because it gives the efficiency also Easy and simple method ,the diode logic uses the fact that diodes conduct only in one direction (they behave like switches).DIODE LOGIC AND =when both are at logic 1 the two diodes are reversed biased and there is no current flowing to ground therefore the output logic is 1 .if one of the input is logic 0 the current will flow through the corresponding diode and through the resistor thus the output logic . DIODE LOGIC OR= if one or both inputs are at logic 1 the current will flow through one or both diodes and output logic is 1.

2. What are the limitations of Diode logic?

The limitations of diode logic is that it use only on the design of the AND and OR logic gates . But the disadvantage is that it cannot perform NOT logic operation .As AND and OR are not complete functions by themselves they cannot perform several logic functions without NOT .

3. What is the need of transistor logic

The transistor logic perform the NOT logic function ,It also perform the several logic gate functions and with the help of the NOT logic easily design not functions of the logic gates. More it boost the signal level.

Lab Task-1

- Design the circuits in MULTISIM according to circuit diagrams given above for all Logic Gates attach the screenshots and verify truth tables.

Lab Task-2: Inverter Gate

- Connect the inverter gate circuit shown in figure.
- Investigate its truth table.
- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of Inverter logic gate.

Vin	0.2	0.8	1.5	1.8	3.5	3.8	4.5	4.8
Vout	5	4.90	3.74	3.87	2.26	2.29	2.24	2.20

Lab Task-2: OR Gate

- Connect the inverter gate circuit shown in figure.
- Investigate its truth table.
- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of Inverter logic gate.

Vin	0.5	1.5	2.5	2.8	3.5	3.8	4.5	4.8	5
Vout	4.99	6.07	1.52	1.58	2,26	2.38	3.56	3.68	5

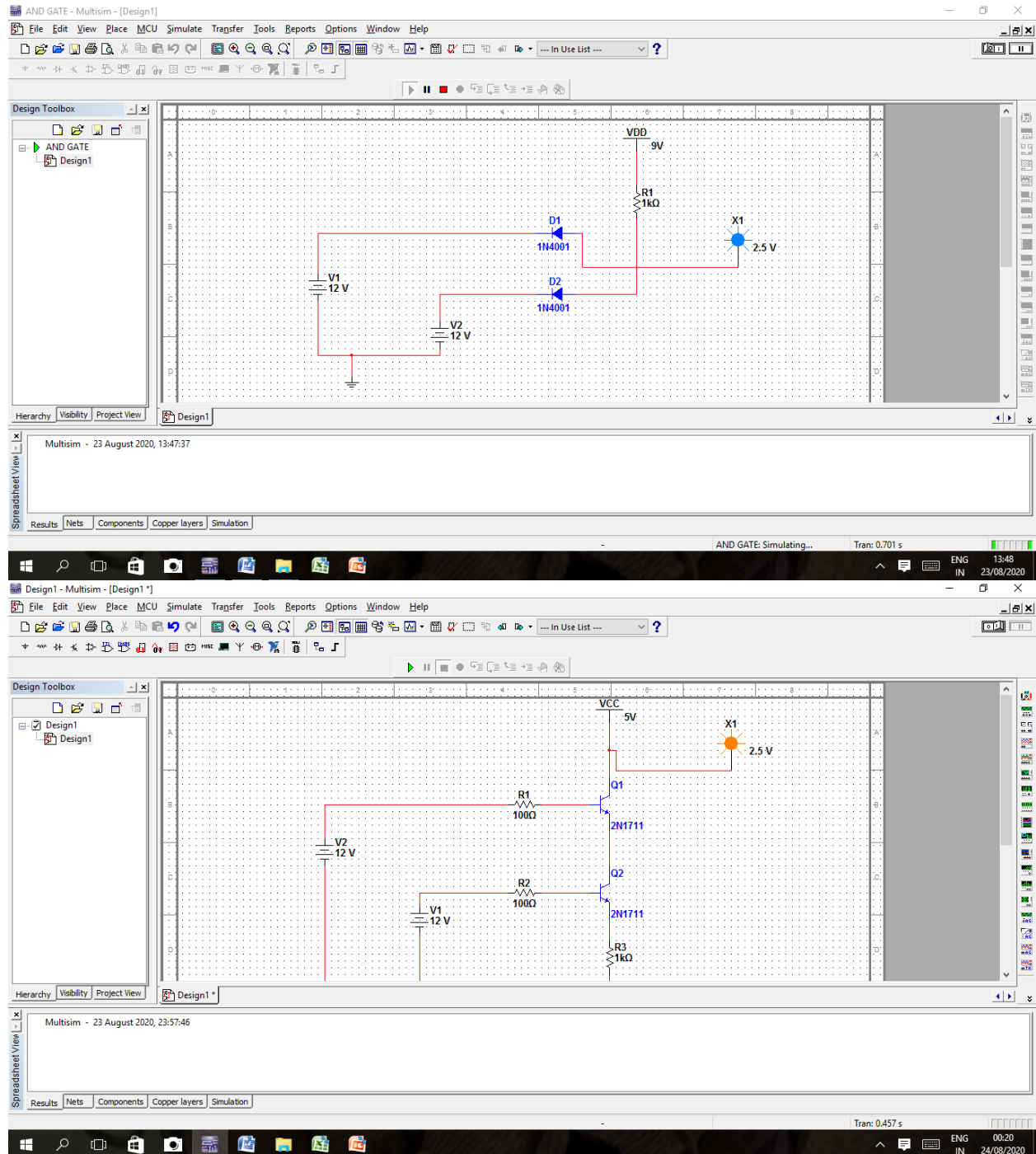
Lab Task-2: AND Gate

- Connect the inverter gate circuit shown in figure.
- Investigate its truth table.
- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of Inverter logic gate.

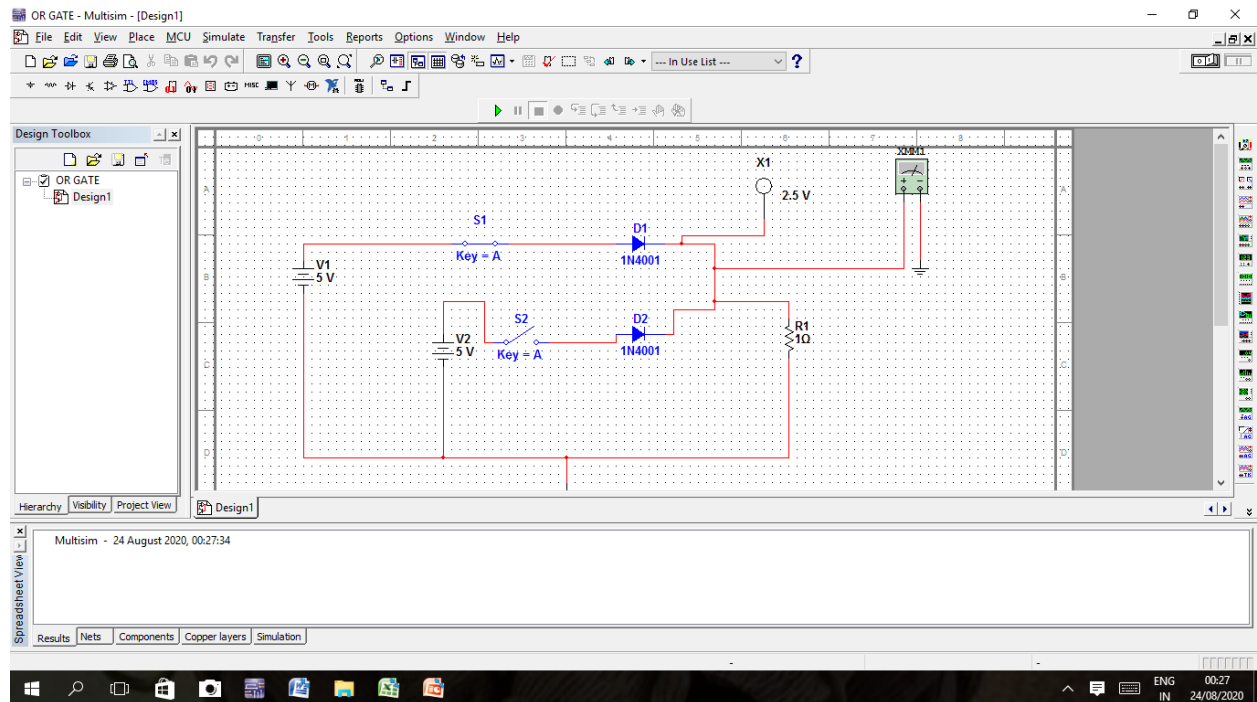
Vin	0.2	0.6	1.5	1.8	2.5	3.4	4.6	4.8	5
Vout	774.1	1.16	1.85	2.20	3.30	3.90	4.81	4.91	5

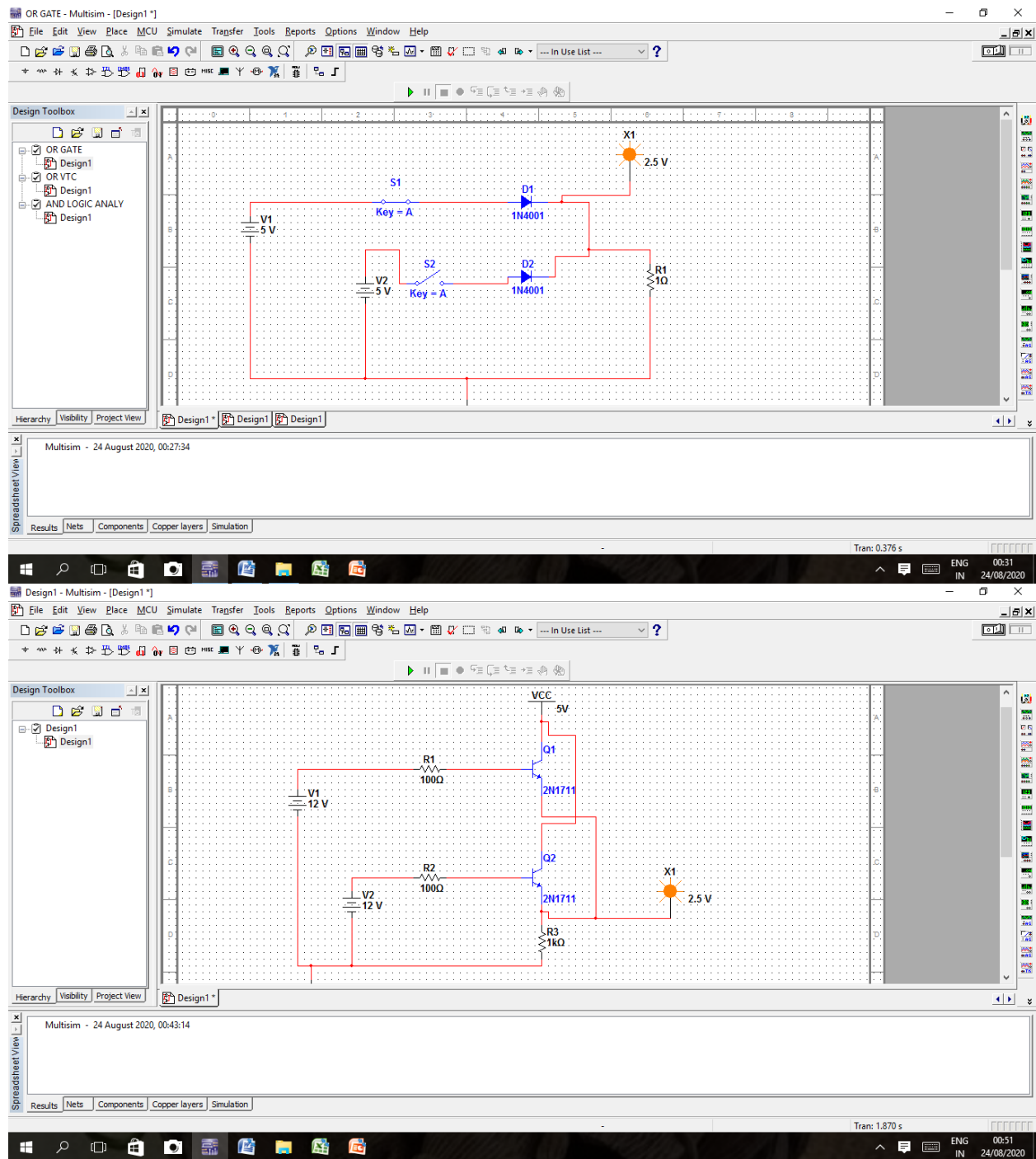
Conclusion: from this lab we designed AND,OR and NOT Logic functions .

AND GATE



OR GATE





NOT GATE

